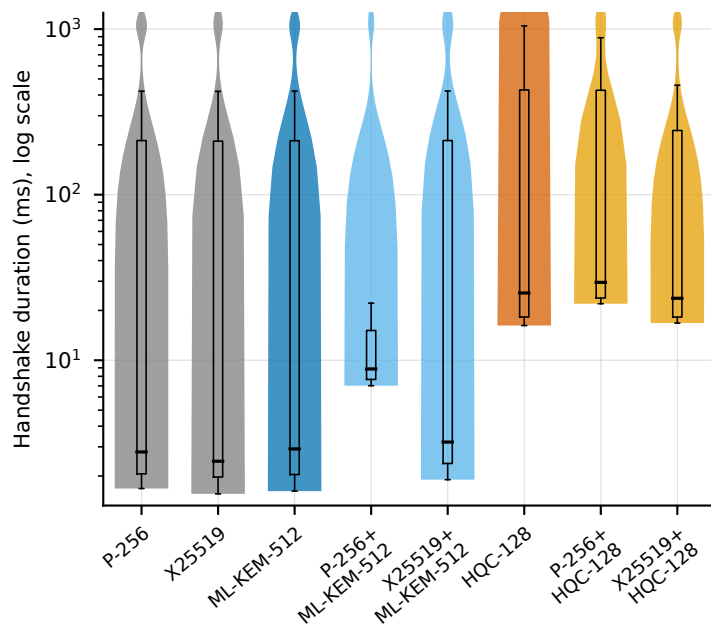
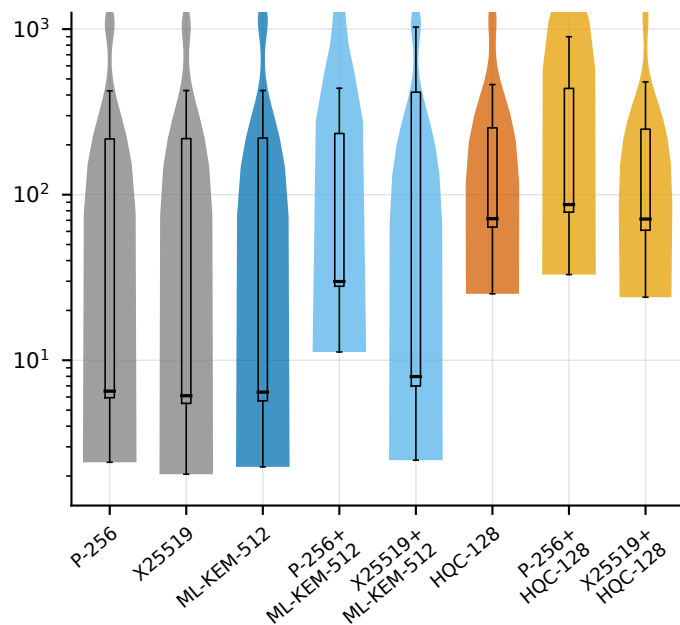


(a)

ed25519 (classical)

**(b)**

mlDSA44 (post-quantum)

**(c)**
 Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlDSA44 Mean (ms)	mlDSA44 CV (%)	mlDSA44 Max (ms)
P-256	295.4	191.2	3.35k	325.7	211.3	7.45k
X25519	276.0	220.2	7.20k	330.8	222.8	7.20k
ML-KEM-512	296.5	215.1	7.19k	346.6	205.9	7.19k
P-256+ML-KEM-512	251.6	220.0	4.09k	421.8	332.3	26.56k
X25519+ML-KEM-512	319.9	231.2	7.60k	353.6	204.1	6.38k
HQC-128	854.6	796.7	109.03k	391.2	209.7	9.12k
P-256+HQC-128	388.5	220.3	12.87k	642.3	532.2	53.39k
X25519+HQC-128	296.0	179.9	3.53k	363.4	199.9	6.44k